

HealthConnect Final Analytics & Decision Support Report

Week 8 – Final Integration, Presentation & Project Showcase

Data Analytics Track | AnalystLab Africa Experience Lab

1. Executive Summary

The HealthConnect Appointment Analytics project examined 5,000 appointment records to support HealthConnect Clinic in understanding missed appointments and improving appointment attendance. The analytical work was developed in Microsoft Power BI using Power Query and DAX, with the final dashboard built around an Overview page and an Attendance Drivers page.

The final validated KPI set consists of a 51.2% no-show rate, 5.3% cancellation rate, 72.7% reminder coverage and 57.8% repeat no-show rate. Week 7 testing confirmed that the main dashboard KPI values remained consistent and that the Year-Month and Appointment Type slicers behaved correctly. The dashboard was refined by adding a Validated Key Findings section and was successfully re-tested.

The final decision-support focus is to communicate the most important attendance patterns clearly, translate them into practical business actions and show how the analytics contribution supports the wider HealthConnect solution. Cross-track integration with Data Science remains to be completed and must be documented truthfully once an actual exchange of outputs takes place.

2. Project Background

HealthConnect Clinic is a fictional healthcare provider seeking to reduce missed appointments, improve appointment attendance, make better use of available appointment slots and provide more effective administrative support to patients. The Data Analytics contribution focuses on transforming appointment data into validated KPIs, attendance insights and decision-support outputs.

The project progressed from initial analysis and dashboard development into testing, refinement and final presentation readiness. Week 8 therefore focuses on finalising the analytical contribution rather than repeating the earlier exploratory analysis.

3. Week 7 → Week 8 Transition

Requirement	Week 8 Transition
Main Week 7 output	Tested and refined HealthConnect Power BI dashboard with KPI validation, slicer testing and documented evidence.
Most important testing result	Core KPI values remained consistent and both key slicers passed functional testing.
Major issue identified	Cross-track validation with Data Science was not completed during Week 7.
Refinement made	Added a Validated Key Findings section to improve dashboard interpretation.

Component considered ready	Power BI dashboard and validated KPI set.
Still needs integration	Analytics findings need to be formally connected to another track, preferably Data Science or Project Management.
Track(s) involved in final integration	Data Science and/or Project Management, depending on the actual exchange completed.
Final presentation focus	Dashboard, validated KPIs, attendance patterns, testing evidence, business recommendations, limitations and overall business value.

4. Final Integration Readiness

Requirement	Response
Final component	HealthConnect Appointment Analytics Power BI dashboard and final decision-support report.
Problem addressed	High levels of missed appointments and the need to identify patterns that can support attendance-focused decisions.
Current status	Tested, refined and ready for final presentation, subject to final cross-track integration evidence.
Week 7 improvement	KPI validation, Year-Month and Appointment Type slicer testing, dashboard refinement and retesting.
Tracks depended on	Data Science for model interpretation and Project Management for wider project integration.
Tracks that may use this output	Data Science and Project Management.
Output provided	Validated KPIs, attendance patterns, dashboard findings and business recommendations.
Output needed from another track	A documented response showing how the analytics findings support model interpretation or project decisions.
Readiness evidence	KPI consistency checks, slicer test screenshots, refinement screenshot, retest evidence and testing record.
Remaining limitations	Cross-track integration is still pending; analysis remains primarily descriptive and does not establish causality.

5. Data and Tools

The final analytical output is based on 5,000 HealthConnect appointment records. Microsoft Power BI was used for dashboard development, Power Query for data transformation and DAX for KPI calculations and analytical measures. The final dashboard includes Year-Month and Appointment Type slicers to support focused exploration of attendance patterns.

6. Final KPIs

KPI	Final Value
No-show rate	51.2%
Cancellation rate	5.3%
Reminder coverage	72.7%

Repeat no-show rate	57.8%
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These KPI values were confirmed during Week 7 testing. The dashboard returned the expected baseline values after slicer selections were cleared, supporting the consistency of the tested calculations.

7. Validated Findings

High no-show rate: The 51.2% no-show rate indicates that missed appointments are a significant operational concern within the analysed dataset.

Repeat no-show behaviour: The 57.8% repeat no-show rate shows that previous attendance history is an important analytical dimension for identifying higher-risk appointment patterns.

Reminder coverage gap: Reminder coverage is 72.7%, meaning a meaningful share of appointments is not covered by the current reminder process.

Attendance varies across appointment characteristics: Earlier analysis identified relevant patterns involving previous no-show history, booking lead time, reminder status, appointment time and appointment type.

Dashboard segmentation supports decision-making: Year-Month and Appointment Type slicers allow decision-makers to move beyond the overall rate and inspect attendance patterns in relevant segments.

8. Business Insights

The final dashboard indicates that HealthConnect should not treat missed appointments as a single undifferentiated problem. Previous no-show behaviour, reminder status, booking lead time, appointment timing and appointment type provide useful analytical dimensions for understanding where attendance risk may be concentrated.

The repeat no-show KPI is particularly useful for decision support because it highlights a patient-history pattern that can be monitored operationally. Reminder coverage also provides a practical process metric that can be improved without requiring a predictive model.

9. Actionable Recommendations

- Prioritise appointments linked to previous no-show history for additional reminder or follow-up attention.
- Improve reminder coverage and monitor whether additional coverage is associated with better attendance outcomes.
- Monitor longer booking lead times and consider additional reminder touchpoints for appointments booked further in advance.
- Use appointment-type segmentation to identify services with stronger no-show patterns and tailor interventions accordingly.
- Review the Power BI dashboard regularly so changes in attendance behaviour can be identified quickly.
- Use validated analytics findings alongside Data Science modelling so descriptive patterns and predictive outputs inform each other.

10. Cross-Track Contribution

The Week 8 assignment requires final integration evidence. At the time of preparing this report, a completed cross-track exchange has not yet been documented. The preferred integration point is Data Analytics → Data Science, where validated no-show patterns can support final model interpretation.

Once the collaboration is completed, replace this paragraph with the actual evidence and document: the track collaborated with, dependency, output provided, output received, integration activity, what changed, how the change improved the solution and the evidence used. Do not claim completion until an actual exchange and use of outputs has occurred.

11. Analytical Limitations

- The analysis is primarily descriptive and does not establish causal relationships.
- Only the tested KPI and slicer scenarios were formally validated in Week 7.
- The project does not yet include a completed cross-track validation record in this report.
- Reminder coverage should not be interpreted as proof that reminders caused improved attendance without further analysis.
- The 5,000-record dataset represents the approved project dataset and should not be assumed to represent all real-world clinic populations.

12. Final Business Value

The final analytics component gives HealthConnect a concise view of appointment attendance performance and the main factors associated with no-show behaviour. It supports operational monitoring, targeted follow-up, reminder-process improvement and data-informed discussion with Data Science and Project Management. The dashboard also provides a professional decision-support output that can be demonstrated during the final project showcase.

13. Conclusion

The Week 8 Data Analytics contribution consolidates the HealthConnect appointment analysis into a tested and presentation-ready decision-support package. The final KPI set remains stable, dashboard filters function as expected and the refined dashboard communicates the key findings clearly. The remaining priority is to complete and document genuine cross-track integration before final submission, then use the final dashboard and report as the basis for the individual presentation.